

**INSTITUTE OF ENGINEERING AND TECHNOLOGY
DEEN DAYAL UPADHYAYA GORAKHPUR UNIVERSITY
GORAKHPUR**



**COURSE STRUCTURE
&
SYLLABUS**

FOR

B.TECH.

in

Artificial Intelligence and Machine Learning

Based

on

AICTE MODEL CURRICULUM

[Effective from the Session: 2025-26]

Semester-II (First Year) Curriculum

S. No.	Type of Course	Course Code	Course Title	Credit
1.	Basic Science Course	ECHE101	Engineering Chemistry	3+0
2.	Basic Science Course	ECHE151	Engineering Chemistry Lab	0+1
3.	Basic Science Course	EMAT104	Mathematics for Machine Learning	3+0
4.	Humanities and Social Sciences including Management Course	HSM101	Professional Communication	3+0
5.	Humanities and Social Sciences including Management Course	HSM151	Professional Communication Lab	0+1
6.	Engineering Science Course	ECE101	Basic Electronics Engineering	3+0
7.	Engineering Science Course	ECE151	Basic Electronics Engineering Lab	0+1
8.	Professional Core Course	AIML102	Conversational AI	3+0
9.	SEC Course	SEC-x	SEC Course	03
10.	AEC Course	AEC-x	AEC Course	02
Total Credit				23



Professional Communication		
Course code	HSM101	
Category	Humanities & Social Sciences Including Management	
Course title	Professional Communication (Theory)	
Scheme and Credits	Credits :3+0	
Pre-requisites (if any)	-	
Unit-1	Fundamentals of Communications: Technical Communication, features, Distinction between General And Technical Communication, Language as a tool of communications, Levels of communication, Interpersonal, Organizational, Mass communication, The flow of communication, Downward, Upward, Lateral/Horizontal (Peer group), Importance of technical communication, Barriers to Communication	9 (Lectures)
Unit-2	Written Communication Words and Phrases, Word formation, Synonyms and Antonyms, Homophones, Select vocabulary of about 500-1000 New words, Correct Usage, all Parts of Speech, Modals, Concord, Articles, Infinitives, Transformation of sentences, Requisites of Sentence Construction, Paragraph Development, Techniques and Methods, Inductive, Deductive, Spatial, Linear, Chronological etc. Business Communication Principles, Sales & Credit letters, Claim and Adjustment Letters, Job Application and Resumes, Reports, Types, Significance, Structure, Style & Writing of Reports. Technical Proposal, Parts, Types, Writing of Proposal, Significance, Negotiation skills.	9 (Lectures)
Unit-3	Presentation Strategies and Soft Skills Nuances and Modes of Delivery, Body Language, Dimensions of Speech, Syllable, Accent, Pitch, Rhythm, Intonation, Paralinguistic features of voice, Interpersonal communication, Definition, Types, Teamwork, Attitude, Way to improve Attitude Listening Skills. Types, Methods for improving Listening Skills.	9 (Lectures)
Unit-4	Value- Based Text Readings Following essays from the prescribed textbook with emphasis on Mechanics of writing. (i) Humanistic and Scientific Approaches to Human Activity by Moody E. Prior (ii) The Language of Literature and Science by A. Huxley (iii) Man and Nature by J. Bronowski (iv) Science and Survival by Barry Commoner (v) The Mother of the Sciences by A.J. Bahm.	9 (Lectures)

Text Book:

1. Improve your Writing ed. V.N. Arora and Laxmi Chandra, Oxford Univ. Press, 2001, New Delhi.
2. Technical Communication- Principles and Practices by Meenakshi Raman & Sangeeta Sharma, Oxford Univ. Press, 2007, New Delhi.
3. Functional skills in Language and Literature, by R.P. Singh, Oxford Univ. Press, 2005, New Delhi

Reference Books:

1. Communication Skills for Engineers and Scientists, Sangeeta Sharma et.al. PHI Learning Pvt. Ltd, 2011, New Delhi.
2. Business Correspondence and Report Writing by Prof. R.C., Sharma & Krishna Mohan, Tata McGraw Hill & Co.Ltd., 2001, New Delhi.
3. Word Power Made Easy by Norman Lewis, W.R. Goyal Pub.& Distributors, 2009, Delhi.
4. Developing Communication skills by Krishna Mohan, MecraBannerji- Macmillan India Ltd. 1990, Delhi.
5. Manual of Practical Communication by L.U.B. Pandey: A.I.T.B.S. Publications India Ltd.; Krishan Nagar, 2013, Delhi
6. English Grammar and Usage by R.P. Sinha, Oxford University Press, 2005, New Delhi.
7. Spoken English- A manual of Speech and Phonetics by R.K. Bansal & J.B. Harrison Orient Blackswan, 2013, New Delhi.

Conversational AI (AIML102)

Credit: 3+0

Syllabus

Course Objective: Students will be able to understand the concepts of chatbot designing and able to build their own chatbots. Students will be able to deploy chatbot for its practical use.

1: Conversational Design Process: Introduction to virtual assistant/chatbot, use cases, what is conversational design, conversational design process, designing conversational flows, writing the script, designing your conversations, Introduction to Dialog Flow, Setting up Dialog Flow.

2: Building blocks of Interaction models: Agents, types of Intents, creating Intents, training phrases, Entities, configuring rich responses, small talk and salutations, Configuring and testing Intents on Google Assistant, Working on Connected Flows.

3: Linear and Non-linear dialogue: Actions & Parameters, understanding slot filling, context, extended Lead Generation, linear dialogue, nonlinear Dialogue, webhook, Fulfilment.

4: Fulfilment: Fulfilment using webhook, basic setup of webhook code, Extracting parameter values and structuring responses, fulfilment using cloud function

5: Deployment: Introduction to Heroku, Deploying to Heroku, Deploying on Alexa, Re-training , Validation & Testing.

Course Outcomes:

C01: Students will understand the concepts of chatbot designing.

C02: Students will be able to build their own chatbots.

C03: Students will be able to deploy chatbot for its practical use.

References

1. Hands-on chatbot with Google Dialogflow, Loonycorn, O'Reilly, Packt publishing.
2. Hands-on chatbots and conversational UI development, Srini Janarthnam, Packt publishing.

Mathematics for Machine Learning (EMAT104)

Credit: 3+0

Course Objective: Students will be able to learn the fundamentals of mathematical concepts used in the area of machine Learning. Students will learn the concept of probability, statistics and linear algebra for machine learning.

Syllabus

1. Introduction to Machine Learning, Introduction to linear equations, Intercepts and slopes, System of equations, Exponentials, radicals and logarithms, Polynomials, Polynomial operations, Factorizations, Introduction to quadratic equations, Quadratic Functions, Maxima and Minima.
2. Probability Basic rules and axioms events, sample space, dependent and independent events, conditional probability, Random variables- continuous and discrete, expectation, variance, distributions- joint and conditional, Bayes' Theorem, Popular distributions- binomial, Bernoulli, poisson, exponential, Gaussian.
3. Statistics Fundamentals of Data: Collection, Summarization, and Visualization; Sampling and Sampling Distributions, Central Limit Theorem; Methods of Estimation, Unbiased estimators; Confidence Interval Estimation: Z-interval, t-interval; Hypothesis Testing, Types of Errors, Rejection Region Approach and p-value Approach.
4. Linear Algebra: Vector space, subspaces, linear dependence and independence of vectors, matrices, projection matrix, orthogonal matrix, idempotent matrix, partition matrix and their properties, quadratic forms, systems of linear equations and solutions; Gaussian elimination, eigenvalues and eigenvectors, determinant, rank, nullity, projections, LU decomposition, singular value decomposition.

Course Outcomes (CO): At the end of this course students will be able to:

CO1: To develop simple algorithms for arithmetic and logical problems. **CO2:** To translate the algorithms to programs & execution (in C language). **CO3:** To implement conditional branching, iteration and recursion.

CO4: To decompose a problem into functions and synthesize a complete program using divide and conquer approach.

CO5: To use arrays, pointers and structures to develop algorithms and programs.

References

1. Christopher M. Bishop. 2006. Pattern Recognition and Machine Learning (Information Science and Statistics). Springer-Verlag New York, Inc., Secaucus, NJ, USA.
2. Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, 2020, Mathematics For Machine Learning. MIT-Press.



Engineering Chemistry Lab		
Course code	ECHE151	
Category	Applied Science Course	
Course title	Engineering Chemistry Lab (Laboratory)	
Scheme and Credits	Credits :0+1	
Pre-requisites (if any)	-	
Exp-1.	Determination of alkalinity in the given water sample.	
Exp-2.	Determination of temporary and permanent hardness in water sample using EDTA.	
Exp-3.	Determination of available chlorine in bleaching powder.	
Exp-4.	Determination of chloride content in water sample.	
Exp-5.	Determination of iron content in the given solution by Mohr's method.	
Exp-6.	pH- metric titration.	
Exp-7.	Determination of viscosity of given liquid.	
Exp-8.	Determination of surface tension of given liquid.	
Exp-9.	Determination of Cell constant and conductance of a solution.	
Exp-10.	Element detection and functional group identification in organic compounds.	
Exp-11.	Preparation of Bakelite and Urea formaldehyde resin.	
Exp-12.	Determination of rate constant of hydrolysis of esters.	



Engineering Chemistry		
Course code	ECHE101	
Category	Applied Science Course	
Course title	Engineering Chemistry (Theory)	
Scheme and Credits	Credits :3+0	
Pre-requisites (if any)	-	
Course Objectives: 1.To teach the fundamental principles and applications of chemistry, including molecular orbital theory, band theory of solids, polymers, electrochemistry and fuels. 2. It seeks to equip students with the knowledge and skills needed for scientific and industrial applications in these areas.		
Unit-1	Molecular orbital theory and its applications to homo-nuclear diatomic molecules. Band theory of solids. Liquid crystals and its applications. Point defects in Solids. Structure and applications of Graphite and Fullerenes. Concepts of nanomaterials and its applications.	9 (Lectures)
Unit-2	Polymers: Basic concepts of polymer-blends and composites. Conducting and biodegradable polymers. Preparations and applications of some industrially important polymers (Buna N, Buna S, Neoprene, Nylon 6, Nylon 6,6 , Terylene). General methods of synthesis of organometallic compound (Grignard Reagent) and their applications in polymerization.	9 (Lectures)
Unit-3	Electrochemistry: Galvanic cell, electrode potential. Lead storage battery. Corrosion, causes and its prevention. Setting and hardening of cement, applications of cement. Plaster of paris. Lubricants- Classification, mechanism and applications. Hardness of water. Disadvantage of hard water. Boiler troubles, Techniques for water softening; Lime-soda, Zeolite, Ion exchange resin, Reverse osmosis. Phase Rule and its application to water system.	9 (Lectures)
Unit-4	Fuels; Classification of fuels. Analysis of Coal. Determination of Calorific Values (bomb calorimeter & Dulong's method). Biogas. Elementary ideas and simple applications of UV, Visible IR and H-NMR spectral Techniques.	9 (Lectures)

Text Books:

1. Chemistry for Engineers, by S. Vairam and Suba Ramesh; Wiley India

Reference Books:

1. Textbook of Engineering Chemistry by Dr. Gopal Krishna Bhatt, Acme Publishers
2. Chemistry (9th ed), by Raymond Chang, Tata McGraw-Hill
3. Chemistry Concepts and Applications by Steven S. Zumdahl; Cengage Learning
4. Engineering Chemistry, Wiley India
5. Engineering Chemistry Author: Abhijit Mallick, Viva Books
6. Text Book of Engineering Chemistry by Harsh Malhotra; Sonali Publications
7. Concise Inorganic Chemistry by J.D. Lee; Wiley India
8. Organic Chemistry (6 ed) by Morrison & Boyd; Pearson Education
9. Physical Chemistry by Gordon M. Barrow; Mc-Graw Hill
10. Organic Chemistry, Volume 1(6 ed)& 2 (5ed) by I. L. Finar; Pearson Education
11. Atkins' Physical Chemistry by Peter Atkins & Julio De Paula; Oxford University Press.
12. Vogel, Authur I, G H. Jeffery, and Arthur I. Vogel. Vogel's Textbook of Quantitative Chemical Analysis. Harlow, Essex, England: Longman Scientific & Technical.

Course Outcomes:

1. Apply their knowledge to analyze and solve problems related to homo – nuclear diatomic molecules, solid – state materials, polymers, electrochemical processes and fuel characteristics.
2. Develop the ability to critically evaluate and make informed decisions regarding the selection and utilization of materials and processes in various scientific and industrial contexts.



Professional Communication Lab		
Course code	HSM151	
Category	Humanities & Social Sciences Including Management	
Course title	Professional Communication Lab (Laboratory)	
Scheme and Credits	Credits :0+1	
Pre-requisites (if any)	-	
1.	Group Discussion: Practical based on Accurate and Current Grammatical Patterns.	
2.	Conversational skills for Interviews under suitable Professional Communication Lab conditions with emphasis on Kinesics.	
3.	Communication Skills for Seminars/Conferences/Workshops with emphasis on Paralinguistic / Kinesics.	
4.	Presentation Skills of Technical Paper/Project Reports/Professional Reports based on proper Stress and Intonation Mechanics.	
5.	Official /Public Speaking based on Rhythmic Patterns.	
6.	Theme-Presentation/Key-Note Presentation based on correct argumentation methodologies.	
7.	Individual Speech Delivery/Conferences with skills to defend Interjections/Quizzes.	
8.	Argumentative Skills/Role Play Presentation with Stress and Intonation.	
9.	Comprehensions Skills based on Reading and Listening Practicals on a model Audio- Visual Usage.	

Reference Books

1. Bansal R.K.& Harrison: Phonetics in English, Orient Longman , New Delhi.
2. Sethi & Dhamija: A Course in Phonetics and Spoken English, Prentice Hall, NewDelhi.
3. L.U.B. Pandey & R.P.Singh, A Manual of Practical Communication, A.I.T.B.S. Pub.India Ltd. Krishan Nagar, Delhi.
4. Joans Daniel, English Pronouncing Dictionary, Cambridge Univ. Press.



Basic Electronics Engineering Lab			
Course code	ECE151		
Category	Basic Engineering Course		
Course title	Basic Electronics Engineering Lab (Laboratory)		
Scheme and Credits	Credits	0+1	
Pre-requisites (if any)	-		
EXP No.	Experiment		
EXP-1	Study of Power Supply, Active and Passive Components, and Bread Board.		
EXP-2	Study of CRO, DSO, Multimeter, and Function generator.		
EXP-3	Study of CRO, Multimeter, and function generator		
EXP-4	To plot V-I Characteristics of PN Junction Diode.		
EXP-5	To plot V-I characteristics of the Zener diode.		
EXP-6	To study the operation of a Half wave rectifier and Measurement of V_{rms} , V_{dc} , and ripple factor		
EXP-7	To study the operation of a Full wave rectifier and Measurement of V_{rms} , V_{dc} , and ripple factor		
EXP-8	To plot the Characteristics of a BJT in a Common Emitter Configuration.		
EXP-9	To plot the Characteristics of a BJT in Common Base Configuration.		
EXP-10	To study Drain Characteristics and Transfer Characteristics of a Junction Fiel Effect Transistor (JFET).		
EXP-11	To study Operational Amplifier as Adder and Subtractor		
EXP-12	To study clipping & clamping circuits.		
<i>Note: Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.</i>			



Detailed B. Tech ECE Curriculum Contents

Basic Electronics Engineering			
Course code	ECE101		
Category	Basic Engineering Course		
Course title	Basic Electronics Engineering (Theory)		
Scheme and Credits	Credits	3+0	
Pre-requisites (if any)	-		
Course Objective: To introduce the basic concept of Electronics engineering to the students			
Unit-1	PN junction diode: Introduction of Semiconductor Materials, Semiconductor Diode, Depletion layer, V-I characteristics, ideal and practical, Transition and Diffusion Capacitance, Diodes breakdown mechanism (Zener and avalanche) Diode Application, Series and Parallel Diode Configuration, Half and Full Wave rectification, Clippers, Clampers, Zener diode as shunt regulator, Voltage-Multiplier Circuits, Light-Emitting Diodes, Liquid-Crystal Displays.		09 (Lectures)
Unit-2	Bipolar Junction Transistors and Field Effect Transistor: Bipolar Junction Transistor, Transistor Construction, Operation, Amplification action. Common Base, Common Emitter, Common Collector Configuration DC Biasing BJTs: Operating Point, Fixed-Bias, Emitter Bias, Voltage-Divider Bias Configuration. Collector Feedback, Emitter-Follower Configuration. Bias Stabilization, CE, CB, CC amplifiers. Field Effect Transistor, Construction and Characteristics of JFETs. MOSFET (Depletion and Enhancement) Type, Transfer Characteristics.		09 (Lectures)
Unit-3	Operational Amplifiers, Introduction and Block diagram of Op Amp, Ideal & Practical characteristics of Op-Amp, Differential amplifier circuits, Practical Op- Amp Circuits (Inverting Amplifier, Noninverting Amplifier, Unity Gain Amplifier, Summing Amplifier, Integrator, Differentiator).		09 (Lectures)
Unit-4	Electronic Instrumentation and Measurements: Digital Voltmeter, Basic Principle and Block Diagram of Oscilloscope, Simple CRO, Measurement of voltage, current phase and frequency using CRO. Introduction of Digital Storage Oscilloscope. Fundamentals of Communication Engineering, Elements of a Communication System, Need of Modulation, Electromagnetic spectrum and typical applications. Introduction of various analog modulation techniques.		09 (Lectures)

Text Books:

1. Robert L. Boylestand / Louis Nashelsky "Electronic Devices and Circuit Theory", Latest Edition, Pearson Education.
2. H S Kalsi, "Electronic Instrumentation", Latest Edition, TMH Publication,.
3. George Kennedy, "Electronic Communication Systems", Latest Edition, TMH,

Reference Books:

1. David A. Bell, "Electronic Devices and Circuits", Latest Edition, Oxford University Press.
2. Jacob Millman, C.C. Halkias, Staya bratajit, "Electronic Devices and Circuits", Latest Edition , TMH.
3. David A. Bell, Electronic Instrumentation and Measurements, Latest Edition, Oxford University Press India.